

Writing Guide

Phase 1: The Project Proposal

The proposal is your "pitch" to get your project topic approved by your supervisor. It is written in the future tense ("The project will...") and typically spans 3 to 5 pages.

Table of Content

1. Abstract
2. Introduction and Background
3. Problem Statement
4. Project Goal and Objectives
5. Proposed System Concept and Architecture
6. Implementation Methodology
7. Scope and Limitations
8. Project Timeline
9. System and Hardware Requirements
10. Expected Outcomes
11. References

- Project Title: A clear, concise title indicating what you are building or researching.
- Introduction / Background: Brief context of the industry or subject area.
- Problem Statement: The exact problem or gap your project aims to solve.
- Objectives: Your specific, measurable targets (e.g., "To design an automated billing system").
- Significance: Why this project matters and who will benefit from it.
- Proposed Methodology: A brief outline of how you intend to do the work, tools you will use, or data you will collect.
- Timeline / Gantt Chart: A visual schedule showing how you will complete the work before the deadline.

Phase 2: The Final Documentation (The 5-Chapter Structure)

Once your project is complete, you will expand your proposal into the final documentation. This is written mostly in the past tense ("The system was tested...") because the work is now finished. [11, 12, 13, 14]



Preliminary Pages (Before Chapter 1) [15]

- Title Page: Project title, your name, department, institution, and date.
- Abstract / Executive Summary: A one-page summary of the entire project, covering the problem, method, results, and conclusion.
- Table of Contents: Page numbers for all chapters and subsections.
- Lists of Tables / Figures: Index of charts, screenshots, or diagrams used. [7, 10, 16, 17, 18]

Chapter 1: Introduction

This sets the foundation of your report. You can adapt much of this from your approved proposal.

- 1.1 Background of the Study: Deep dive into the history and context of the topic.
- 1.2 Problem Statement: Clear description of the exact pain point or research gap.
- 1.3 Aim and Objectives: The primary goal (aim) and specific steps to reach it (objectives).
- 1.4 Significance of the Study: The practical benefits or academic contributions of your work.
- 1.5 Scope and Limitations: What your project covers, and what it intentionally leaves out.

Chapter 2: Literature Review

This section proves you understand what already exists in your field. [10, 20]

- 2.1 Conceptual Review: Defining terms and core concepts related to your project.
- 2.2 Theoretical Framework: The existing theories or engineering methodologies supporting your approach.
- 2.3 Review of Related Work: Summaries of similar projects, software, or studies built by others, highlighting their strengths and weaknesses.
- 2.4 Gap in Literature/Existing Systems: Explanation of how your project improves upon or fills the gaps of those past works. [10, 21, 22, 23, 24]

Chapter 3: Methodology & System Design

The "how-to" section. It must contain enough detail for someone else to replicate your work. [10, 25, 26]

- For Research Projects: Describe your research design, data collection instruments (like surveys or interviews), and data analysis methods. [1, 10]
- For Software/Engineering Projects: Describe your system architecture, data flow diagrams (DFDs), database designs (ER diagrams), and user interface mockups. [27]

Chapter 4: Implementation, Testing, & Results [28]

This is the core of what you actually accomplished. [29, 30]

- 4.1 System Implementation / Data Presentation: Screenshots of your working system or the raw findings from your data analysis.
- 4.2 Testing / Analysis: How you tested the project (e.g., unit testing, user acceptance testing) and the performance outcomes.
- 4.3 Discussion of Results: What the results mean and how they successfully map back to your Chapter 1 objectives. [17, 29]

Chapter 5: Conclusion & Recommendations and Future Work

Wrap up your project and look toward the future.

- 5.1 Summary: A brief recap of the problem, your solution, and what was achieved.
- 5.2 Conclusion: Final thoughts on whether the project succeeded or failed to meet expectations.
- 5.3 Recommendations / Future Work: Ideas on how future students or developers can expand or improve your project.

End Matter (After Chapter 5)

- References / Bibliography: A list of all books, papers, and websites you cited, formatted in your school's required style (usually APA, IEEE, or MLA). [1, 7, 31, 32, 33]
- Appendices: Large chunks of data that would disrupt the text flow, such as full source code, sample survey questionnaires, or extensive data tables. [34, 35, 36]

Report

Which includes all the chapters